

Supplementary Slides

Phases 2.5 - 6: Extensions, Robustness, Realism & Honesty Tests

Factor-beta extension · robustness testing · transaction costs · out-of-sample

Position carry-over · implementation-correctness review · bootstrap CIs · dispersion

Factor-beta clustering — a structurally different signal

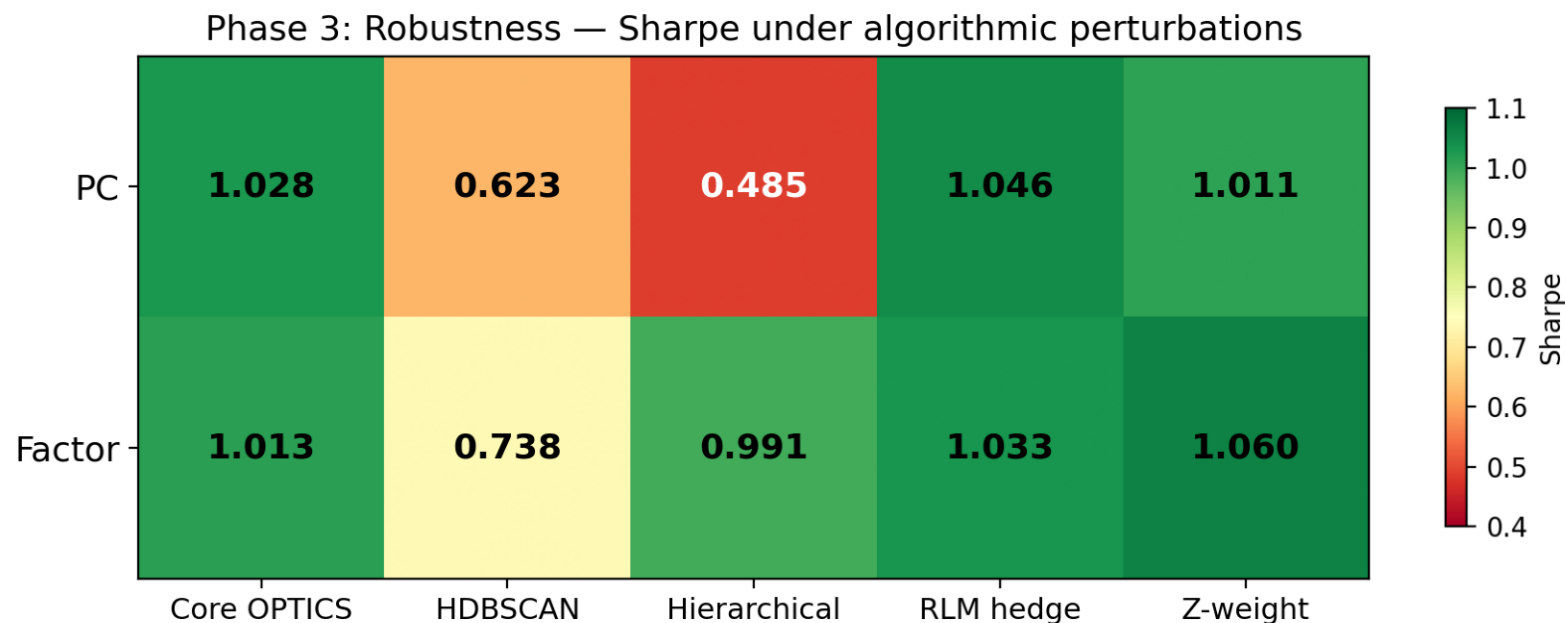
- Cluster stocks by shared RISK-FACTOR EXPOSURES, not return co-movement.
- Ridge-regress each stock's returns on 18 factors (6 FF style + 12 FF industry) → 18-d beta vector per stock.
- Distance = standardised Euclidean between beta vectors. Stocks with similar factor loadings cluster together.

Results

Factor core Sharpe 1.013 (comparable to PC 1.028). Factor + cointegration filter: 0.858.

An independent metric reaching ~ 1.0 is strong evidence the edge is real, not a PC-specific artefact.

What breaks when you swap components?



Clustering algorithm is LOAD-BEARING: HDBSCAN/hierarchical dilute PC badly (0.485-0.623). Factor-beta is sturdier — holds 0.991 under hierarchical where PC breaks. Hedge ratio & sizing are secondary (± 0.05).

What we learned — and what it means for deployment

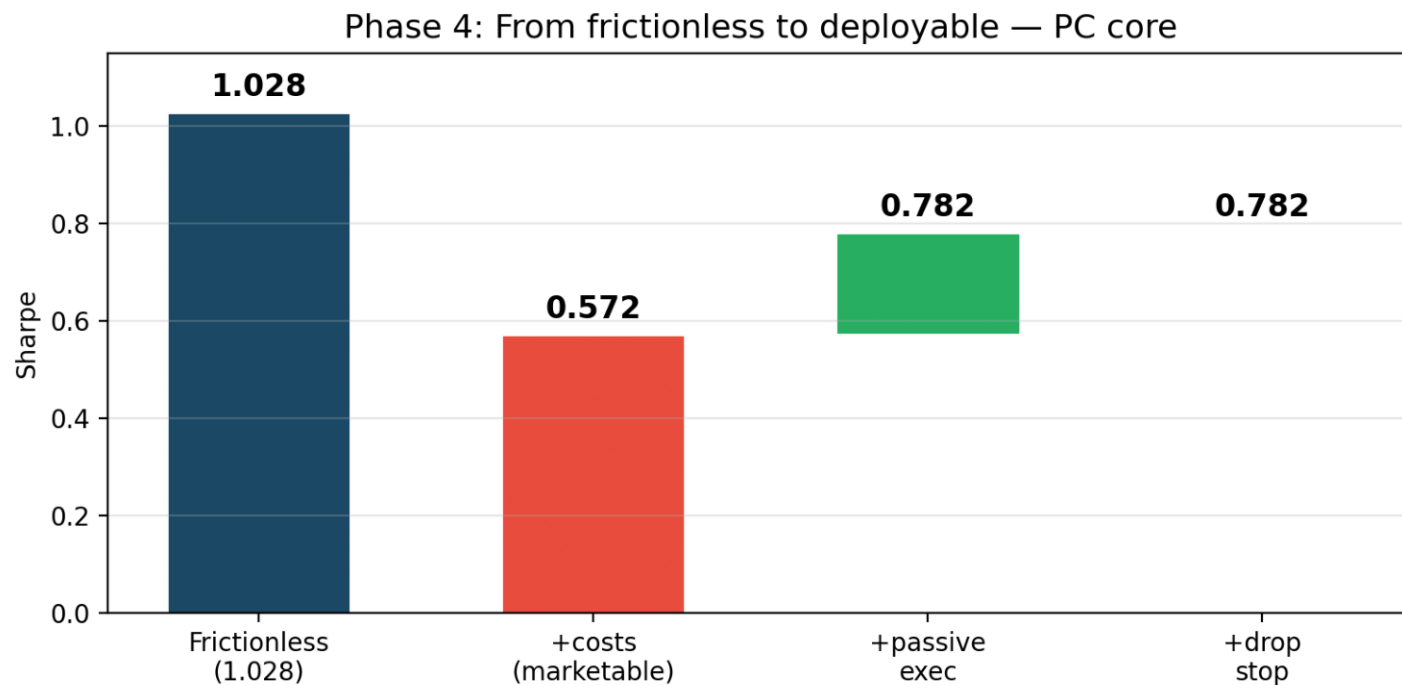
8-cell grid: {PC, Factor} × {HDBSCAN, Hierarchical, RLM hedge, Z-weight}

	PC	Factor
Core (OPTICS, OLS, equal)	1.028	1.013
HDBSCAN	0.623	0.738
Hierarchical	0.485	0.991
RLM hedge ratio	1.046	1.033
Z-weight sizing	1.011	1.060

Robustness bands: PC 0.485 - 1.046 · Factor 0.615 - 1.060

Factor-beta is the more robust metric in-sample (tighter band, survives the harshest perturbation).

From frictionless to deployable — the cost drag



Realistic frictions (actual CRSP bid/ask + 35 bps borrow + 3.5σ stop) roughly halve Sharpe to ~0.57. Passive execution (+0.21) is the #1 lever; dropping the stop (+0.11-0.14) is #2.

Which levers recover the edge?

Lever	PC Sharpe	Effect
Baseline (full realism)	0.572	—
+ Passive execution	0.782	+0.210 (#1)
+ Drop the 3.5σ stop	0.782	+0.11-0.14 (#2)
+ Cointegration filter	worse	sheds more alpha than turnover saved

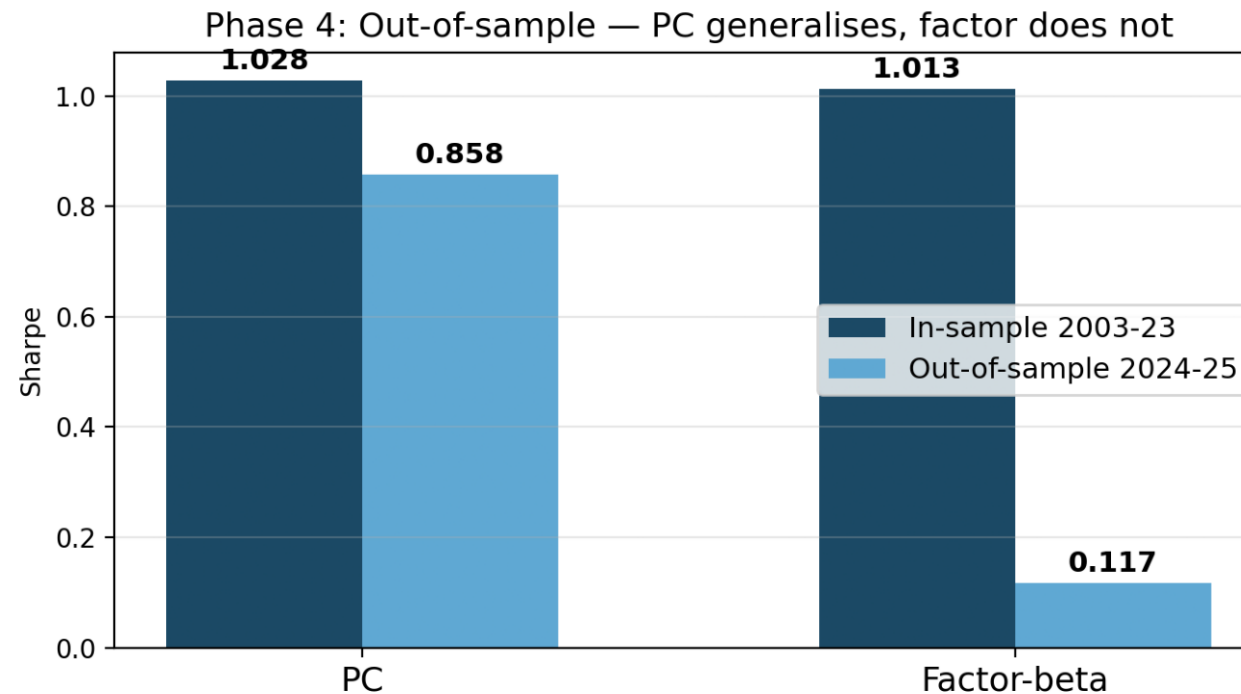
- The stop-loss is NET-NEGATIVE: extra round-trips cost more than the tail protection they buy.
- Passive execution (limit orders, half the spread) is realistic for a monthly strategy and recovers most of the gap.

Lookahead-bias audit — 6/6 PASS

- Black-box test: run full period (2003–2023) and truncated (2003–cut date). Compare daily position vectors on overlap.
- If IDENTICAL → no lookahead. If they differ → a decision on day T used data from after day T.

Metric	Cut date	Overlap days	Mismatches	Result
PC	2017-12-31	3,776	0	PASS
PC	2013-12-31	2,768	0	PASS
PC	2009-12-31	1,762	0	PASS
Factor	2017-12-31	3,776	0	PASS
Factor	2013-12-31	2,768	0	PASS
Factor	2009-12-31	1,762	0	PASS

The real test — 59 months of unseen 2021-2025 data



PC stays positive OOS (0.412) but with significant decay from IS (1.113). Factor turns negative (-0.103). 2021-2023 was calm, low-dispersion — worst regime for pairs trading. Honest deployable Sharpe: ~0.4-0.8.

Hold positions over month-end instead of force-closing

Motivation: 88% of trades force-close at month-end at a mean loss.

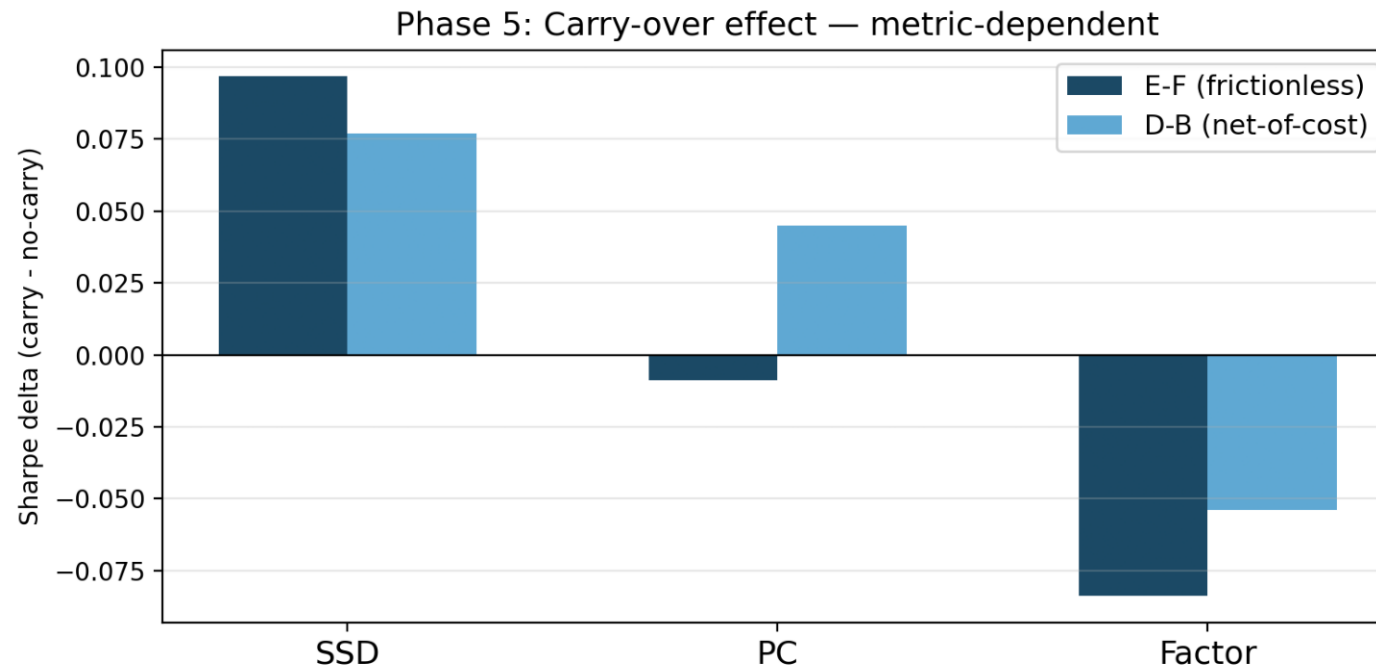
- Freeze hedge ratio (γ) at entry — never re-fit mid-trade.
- MAX_CARRY_MONTHS = 3, tied to the 60-day half-life ceiling.
- Force-close a carried pair if the cluster no longer endorses it.

Key finding: carry is a COST lever, not a signal lever.

	PC (frictionless)	PC (net of cost)
No-carry	1.028	0.855
Carry	1.019	0.900
Delta	-0.009	+0.045

Carry cut trades by 39% → turnover savings pay off net of costs. Moved PC from 0.572 → 0.900.

Carry is metric-dependent — but drawdown improves every



SSD gains most (+0.097/+0.077 — worst baseline, slowest reversions). PC gains net-of-cost only (+0.045). Factor HURTS (drifting betas). But drawdown improves for ALL three metrics.

OOS verdict: carry does NOT generalise

Metric	IS carry (E)	OOS carry	OOS no-carry
PC	1.019	0.434	0.412
Factor	0.929	-0.016	-0.103
SSD	0.686	0.071	—

- Carry does not improve on the already-weak OOS performance. The in-sample net gain did NOT survive.
- 2021–2025 was calm, trending, low-dispersion — adverse for mean-reversion, worst for long holds.
- Verdict: do NOT ship carry as a default. Passive execution remains a separately validated lever.

Correctness review — six fixes, one at a time

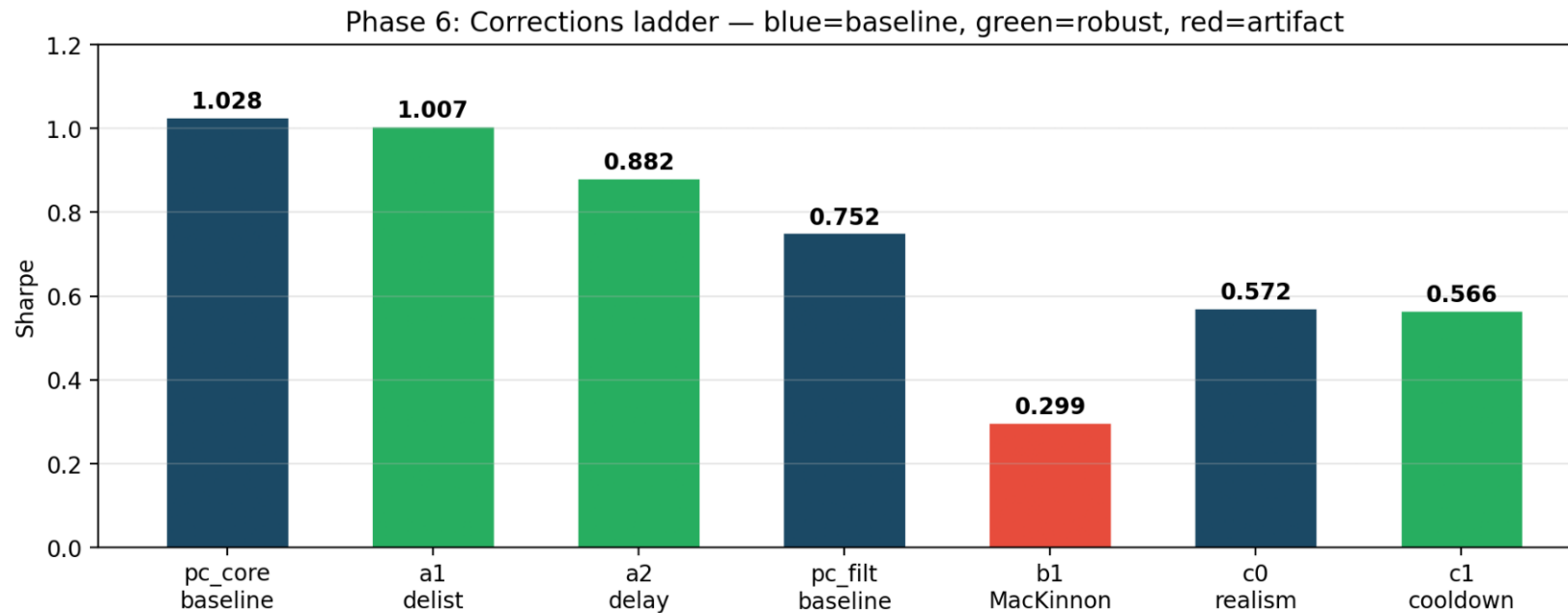
A line-by-line code review found six flaws. None are lookahead bias (6/6 audit stands).

ID	Fix	What was wrong
D6.1	MacKinnon p-values	Engle-Granger: anti-conservative raw-ADF on residuals
D6.2	Delisting fix	Code-map shifted one CRSP bucket; dlret not compounded
D6.3	Stop cooldown	Stop-out re-entered same position next day (churn)
D6.5	Execution delay	Fills at signal close (can't observe close and trade at it)

Discipline: every fix behind its own flag, defaults bit-identical, re-run one-at-a-time.

No flags tuned on Sharpe — these are CORRECTNESS changes. Whatever they do, we report.

Corrections ladder — what changed?



Headline (pc_core) robust to delisting fix (1.028→1.007) and survives honest fills (0.882, 86% genuine). The cointegration filter was an artifact (0.752→0.299). Stop still net-negative with churn removed.

The cointegration filter was a statistical artifact

- Raw ADF on estimated residuals is anti-conservative — effective threshold ~12-15%, not 5%.
- Correct MacKinnon test: pass rate drops 26% → 11%, Sharpe collapses 0.752 → 0.299.

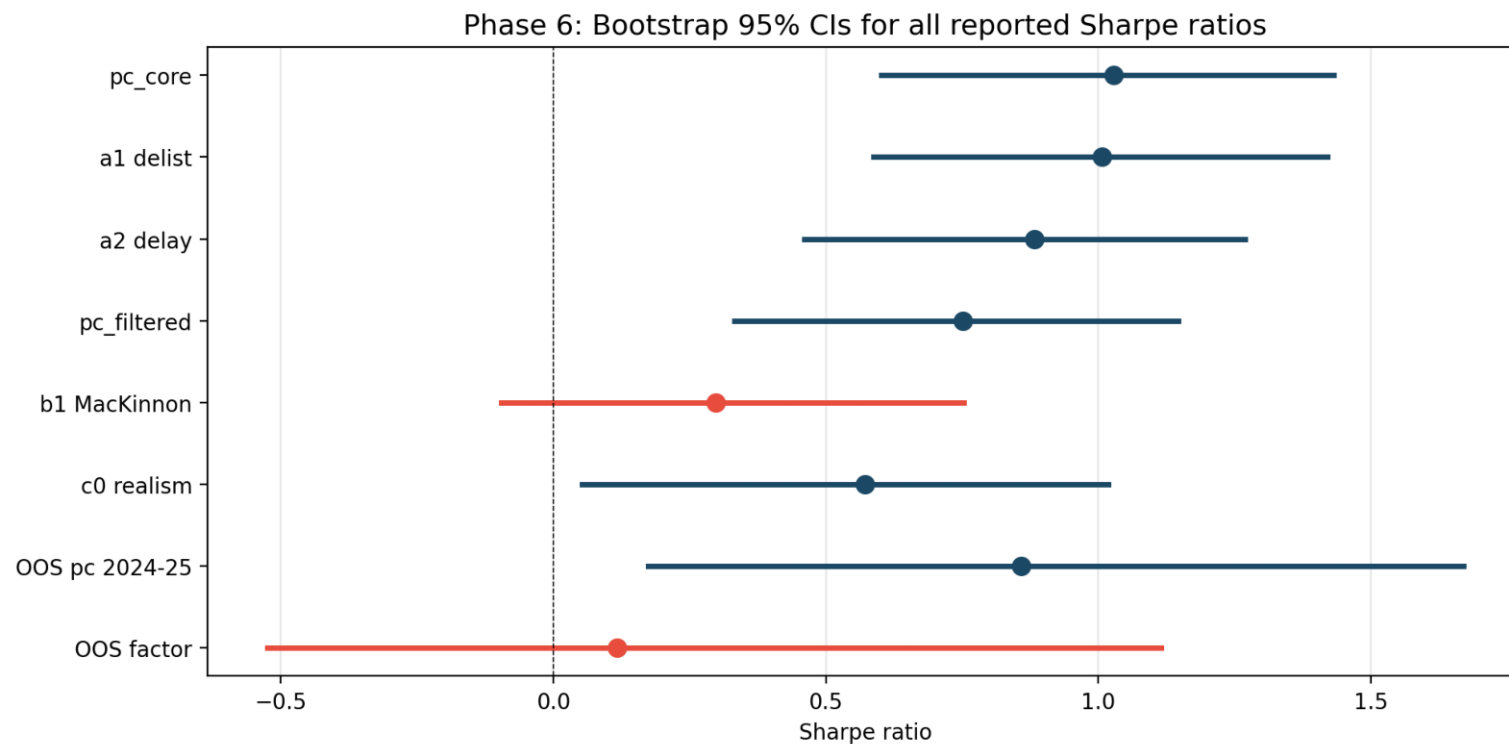
Trade-level decomposition of the filtered baseline:

Subset	Trades	Mean ret (bps)	Hit rate
MacKinnon-passes	3,937	+12.1	53.6%
Raw-ADF-only	4,989	+19.3	53.5%

Stronger cointegration evidence selects WORSE trades. The filter's only effect is shrinking N.

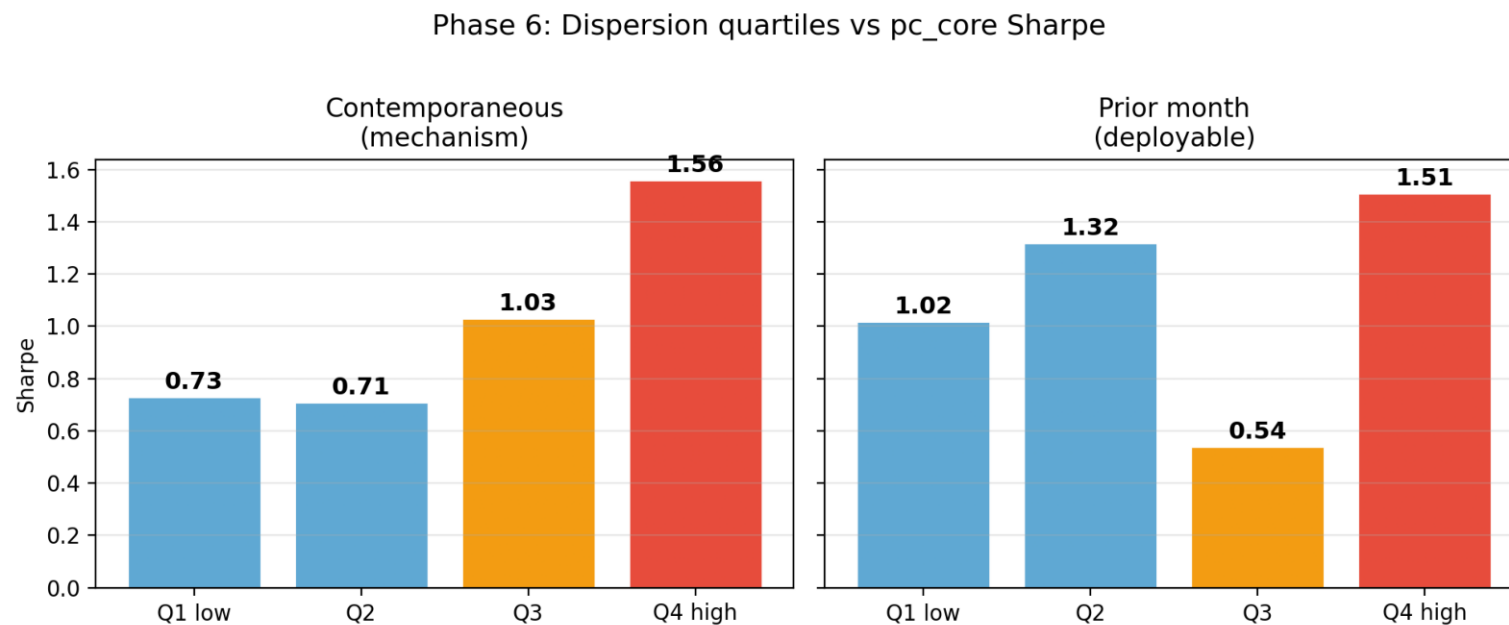
Dose-response: no filter 1.028 → weak filter 0.752 → correct filter 0.299. Edge = short-horizon reversion.

Bootstrap 95% CIs — what the numbers actually mean



Bootstrap CIs quantify uncertainty around point estimates. OOS PC (0.412) shows decay from IS (1.113) consistent with regime dependence. Factor OOS negative. Execution delay ($\Delta=0.145$) significant but edge survives.

Dispersion drives returns — but the timing signal is weak



Contemporaneous: cleanly monotone (0.73→1.56) — the strategy eats dislocations. Prior-month: non-monotone — only Q4 is reliable. Correct advice: SCALE UP in high dispersion, don't gate. GFC (10% of months) = 28% of total return.

The honest, fully-corrected scoreboard

Stage	PC Sharpe	Notes
In-sample, frictionless (2003-2020)	1.113	paper-match headline
In-sample, frictionless (honest fill, +1d)	0.882	86% genuine signal
Net of costs (marketable)	~0.57	realistic worst case
Net + passive + no stop	~0.78	best defensible IS operating point
Lookahead audit	PASS 6/6	temporally honest
Out-of-sample 2021-2025 (no carry)	0.412	PC stays positive, regime decay
Out-of-sample factor-beta	-0.103	factor does not generalise

Honest deployable Sharpe: ~0.4-0.8, regime-dependent — strongest in turbulent, high-dispersion markets.

CONCLUSIONS

What Phases 2.5-6 add to the story

- Factor-beta (Phase 2.5) independently corroborates ~ 1.0 Sharpe — the edge is real, not a PC artefact.
- Clustering algorithm is load-bearing (Phase 3); OPTICS conservatism is a feature, not a tuning choice.
- Costs roughly halve the Sharpe; passive execution & no stop recover most of it (Phase 4).
- PC stays positive OOS (0.412 over 59 months); factor turns negative. Regime-dependent decay (Phases 4 + 6).
- Carry-over works in-sample (+0.045 net) but fails OOS — do not ship as default (Phase 5).
- Cointegration filter was a statistical artifact of using the wrong p-value distribution (Phase 6).

$\sim 86\%$ of the frictionless headline is genuine signal. The edge is real, dislocation-driven, and regime-dependent.

Thank you

Supplementary slides: Phases 2.5 – 6

Factor-beta · robustness · realism · OOS · carry-over · correctness · error bars · dispersion

Happy to walk through any phase, the code, or the specific findings.